

Female reproductive tract module validation report

Metronidazole simulations

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i Note

This report has been created by running the simulations. The results have been stored in folder ' C:/Projects/Female-Reproductive-Tract-PBPK-Module/Reports/Results/SimulationResults/2025-10-06 17-25 '.

1 Introduction

This report presents the validation of the female reproductive tract module using Metronidazole simulations. The module is designed to simulate the pharmacokinetics of drug targetting tissues in the female reproductive tract, and distribution of the drug upon local administration (e.g., vaginal, intrauterine or cervical).

Metronidazole is an antibiotic and antiprotozoal medication used to treat a range of infections, including bacterial vaginosis, trichomoniasis and endometritis. A PBPK model for IV administration of metronidazole was obtained from literature ([Dallmann et al. 2018](#)), and extended for oral and vaginal administration.

2 Compound: Metronidazole

2.1 Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	9500 mg/l		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	-0.02 Log Units		Measurement	True
Fraction unbound (plasma, reference value)	0.89		Measurement	True
Specific intestinal permeability (transcellular)	4.32E-06 cm/min	Parameter Identification	Fit	True
Is small molecule	Yes			
Molecular weight	171 g/mol			
Plasma protein binding partner	Albumin			

2.2 Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

2.3 Processes

2.3.1 Systemic Process: Total Hepatic Clearance-CL_{hep}

Species: Human

2.3.1.1 Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.89	
Lipophilicity (experiment)	0 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	0.013 1/min	

2.3.2 Metabolizing Enzyme: CYP3A4-Pearce 2013

Molecule: CYP3A4

2.3.2.1 Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0.72 nmol/min/mg mic. protein	
Km	33.1 mmol/l	
kcat	72.46 1/min	

2.3.3 Metabolizing Enzyme: CYP2A6-Pearce 2013

Molecule: CYP2A6

2.3.3.1 Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0.213 nmol/min/mg mic. protein	
Km	0.383 mmol/l	
kcat	1.45 1/min	

2.3.4 Systemic Process: Renal Clearances-CLren

Species: Human

Name	Value	Value Origin
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2.3.4.1 Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.89	
Plasma clearance	0.15 ml/min/kg	
Specific clearance	0.022 l/min	

3 Clinical Data

Data from the following studies has been used for model validation Fredricsson et al. (1987):

Reference	Administration_route	Dose	Clinical_data	Population	Weight_kg	Height_cm	Age_years	Number_of_participants
Männistö 1984	IV (20 min)	500 mg	Plasma; Endometrium; Myometrium; Ovarium; Tuba	European ICRP (female)	64.6	160.0	48.9	20
Männistö 1984	PO	400 mg TID	Plasma; Endometrium; Myometrium; Ovarium; Tuba	European ICRP (female)	64.8	163.1	47.5	10
Salas-Herrera 1991a	PO	400 mg BID	Plasma; Cervicovaginal fluid	European ICRP (female)	62.8	NA	25.3	6
Cunningham 1994	PO	500 mg	Plasma	European ICRP (female)	59.1	NA	27.4	12
Cunningham 1994	Vaginal gel	37.5 mg	Plasma	European ICRP (female)	59.1	NA	27.4	12
Fredericsson 1987	IV (15 min)	500 mg	Plasma	European ICRP (female)	22.5	NA	NA	5
Fredericsson 1987	PO	500 mg	Plasma	European ICRP (female)	22.5	NA	NA	5
Fredericsson 1987	Vaginal tablet	500 mg	Plasma	European ICRP (female)	22.5	NA	NA	5
Salas-Herrera 1991b	Vaginal pessary	500 mg BID	Plasma	European ICRP (female)	60.2	NA	25.3	12
Mattila 1983	IV (20 min)	500 mg	Plasma	European ICRP (male)	77.0	180.0	22.0	6
Mattila 1983	PO	500 mg	Plasma	European ICRP (1 male / 8 females)	77.0	178.0	22.0	9
Mattila 1983	Vaginal	500 mg	Plasma	European ICRP (female)	58.0	165.0	23.0	6

4 Results

Comparison of individual simulations with observed data are presented in the following figures.

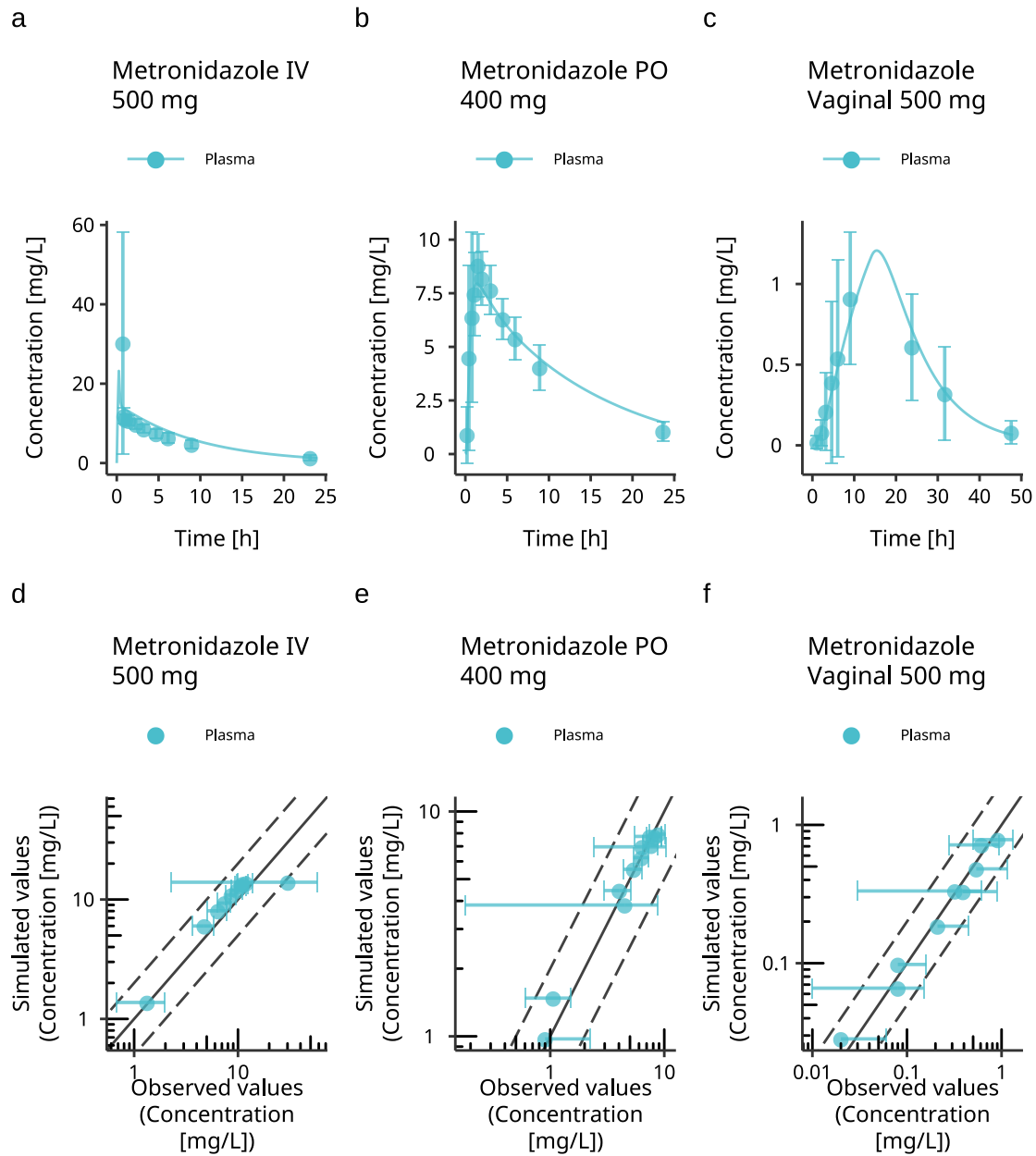


Figure 1: Concentration-time profiles (a, b, c) and goodness-of-fit plots (d, e, f) of metronidazole administered as 500 mg intravenous (a, d), 400 mg oral (b, e), and 500 mg vaginal (c, f) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Fredericsson et al. 1987). Solid line represent unity, and dashed lines represent 2-fold deviation.

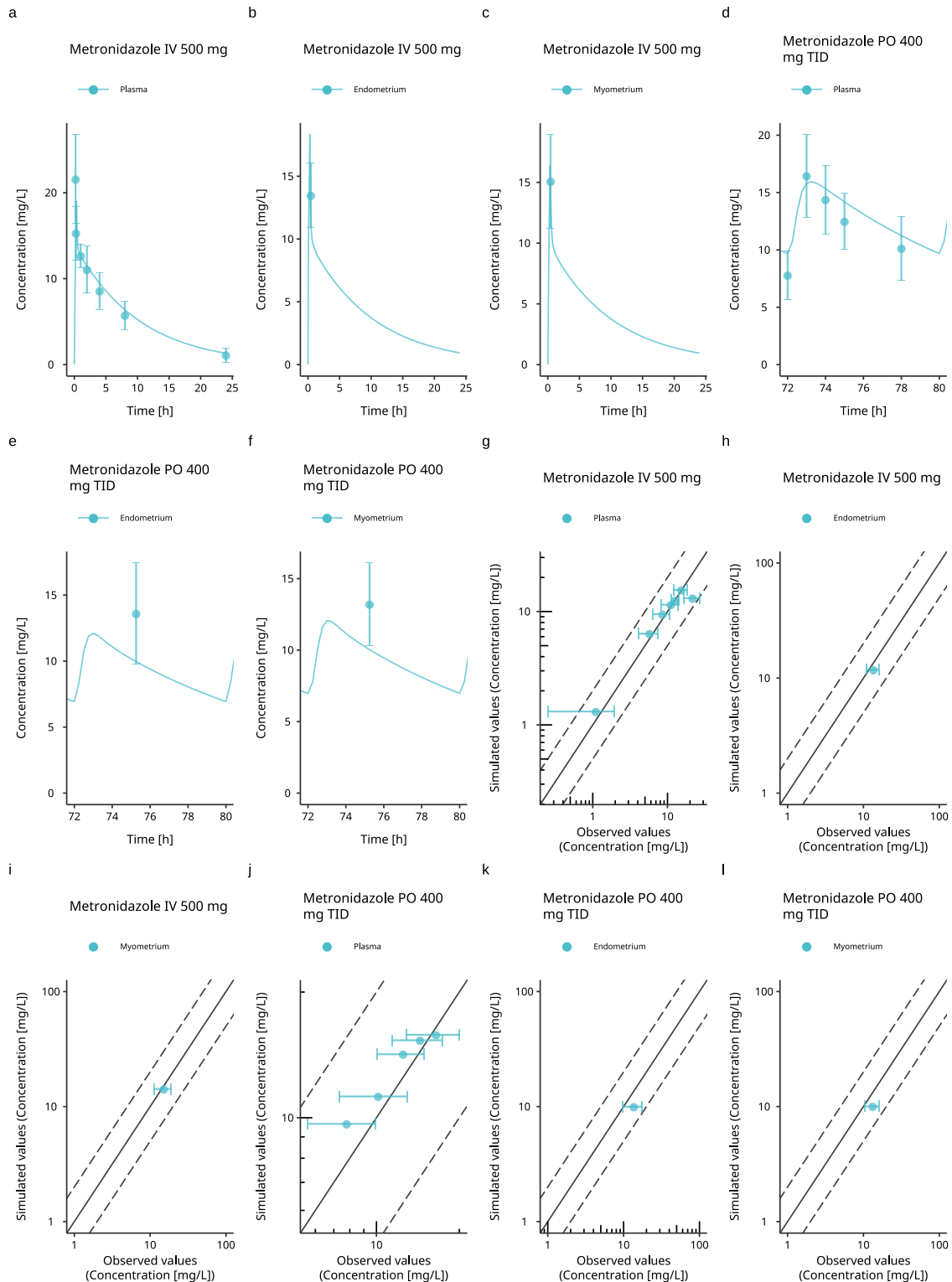


Figure 2: Concentration-time profiles (a, b, c, d, e, f) and goodness-of-fit plots (g, h, i, j, k, l) of metronidazole administered as 500 mg intravenous (a, c), and 400 mg thrice daily oral (b, d) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Männistö et al. 1984). Solid line represent unity, and dashed lines represent 2-fold deviation.

Mattila 1983

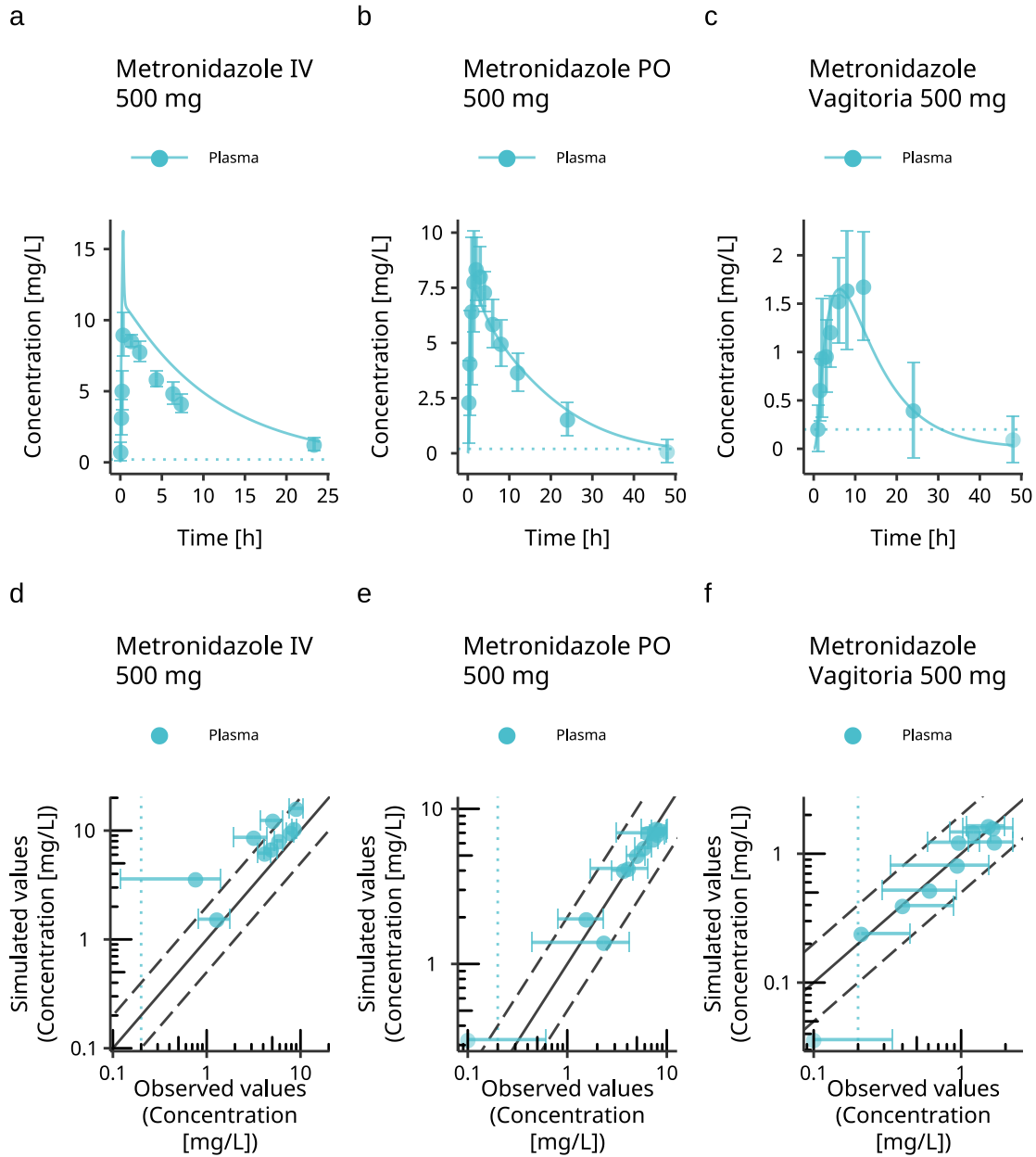


Figure 3: Concentration-time profiles (a, b, c) and goodness-of-fit plots (d, e, f) of metronidazole administered as 500 mg intravenous (a, d), 500 mg oral (b, e), and 500 mg vaginal (c, f) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Mattila et al. 1983). Solid line represent unity, and dashed lines represent 2-fold deviation.

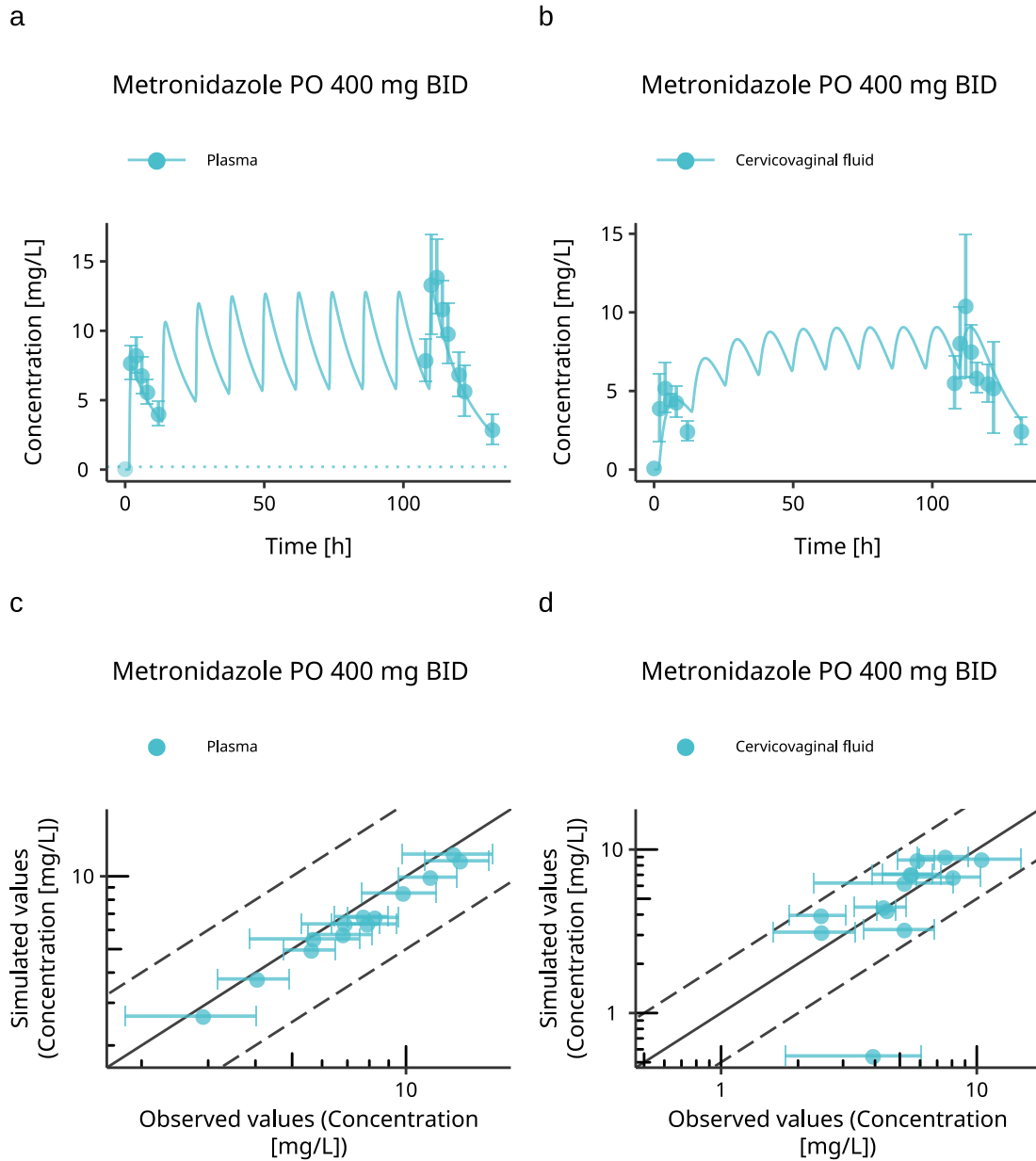


Figure 4: Concentration-time profiles (a, b) and goodness-of-fit plots (c, d) of metronidazole administered as 400 mg bidaily oral (a, b) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Salas-Herrera et al. 1991). Solid line represent unity, and dashed lines represent 2-fold deviation.

Cunningham 1994

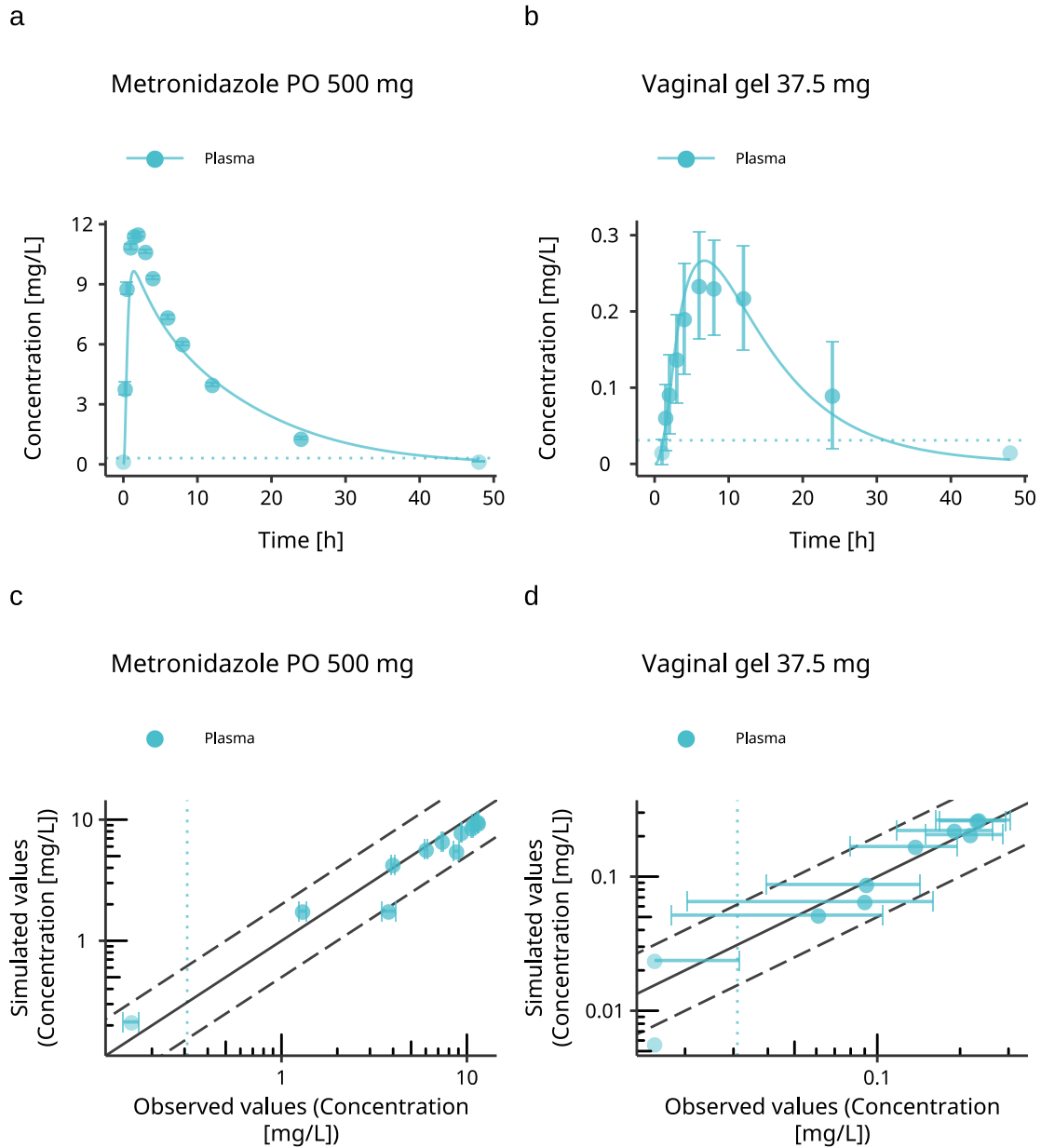


Figure 5: Concentration-time profiles (a, b) and goodness-of-fit plots (c, d) of metronidazole administered as 500 mg oral (a, c), and 37.5 mg vaginal (b, d) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Cunningham et al. 1994). Solid line represent unity, and dashed lines represent 2-fold deviation.

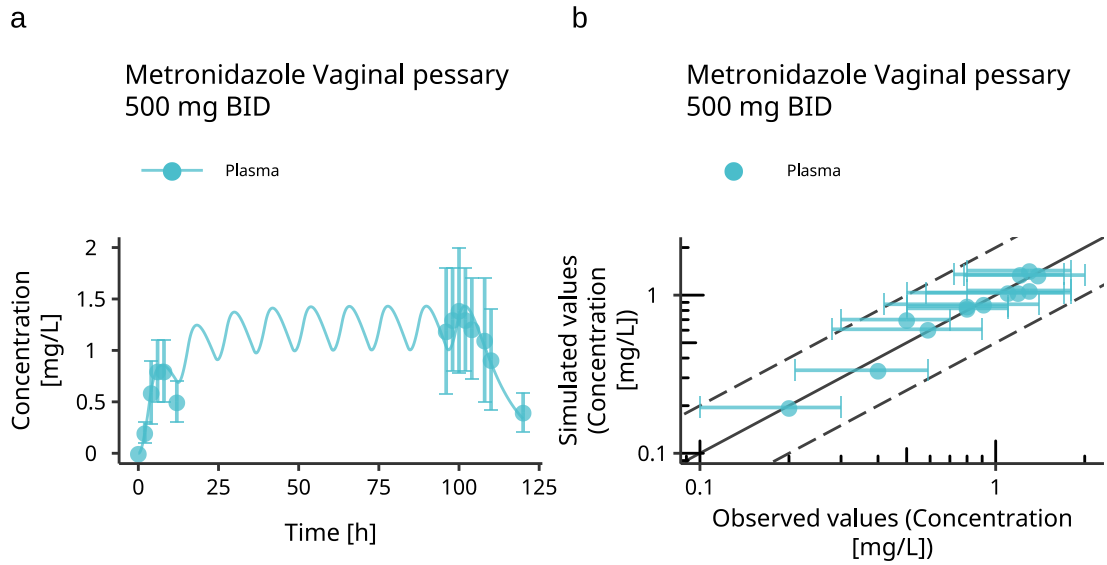


Figure 6: Concentration-time profiles (a) and goodness-of-fit plots (b) of metronidazole administered 500 mg bidaily vaginal (a, b) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Salas-Herrera et al. 1991). Solid line represent unity, and dashed lines represent 2-fold deviation.

Goodness-of-fit visual diagnostic plots for the PBPK model performance for intravenous, oral and vaginal administration are presented in the following figures.

Metronidazole intravenous administration

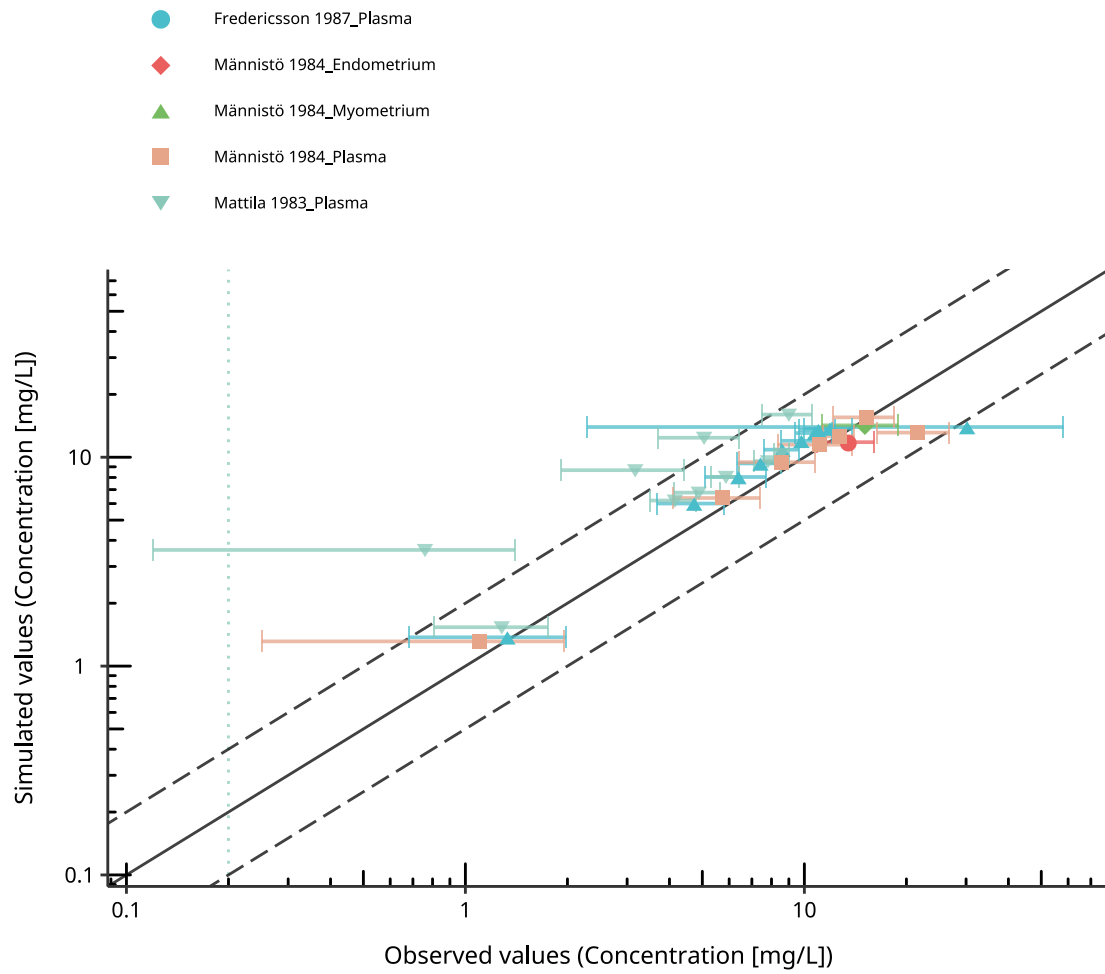


Figure 7: Predicted versus observed plot for oral administration of metronidazole.

Metronidazole oral administration

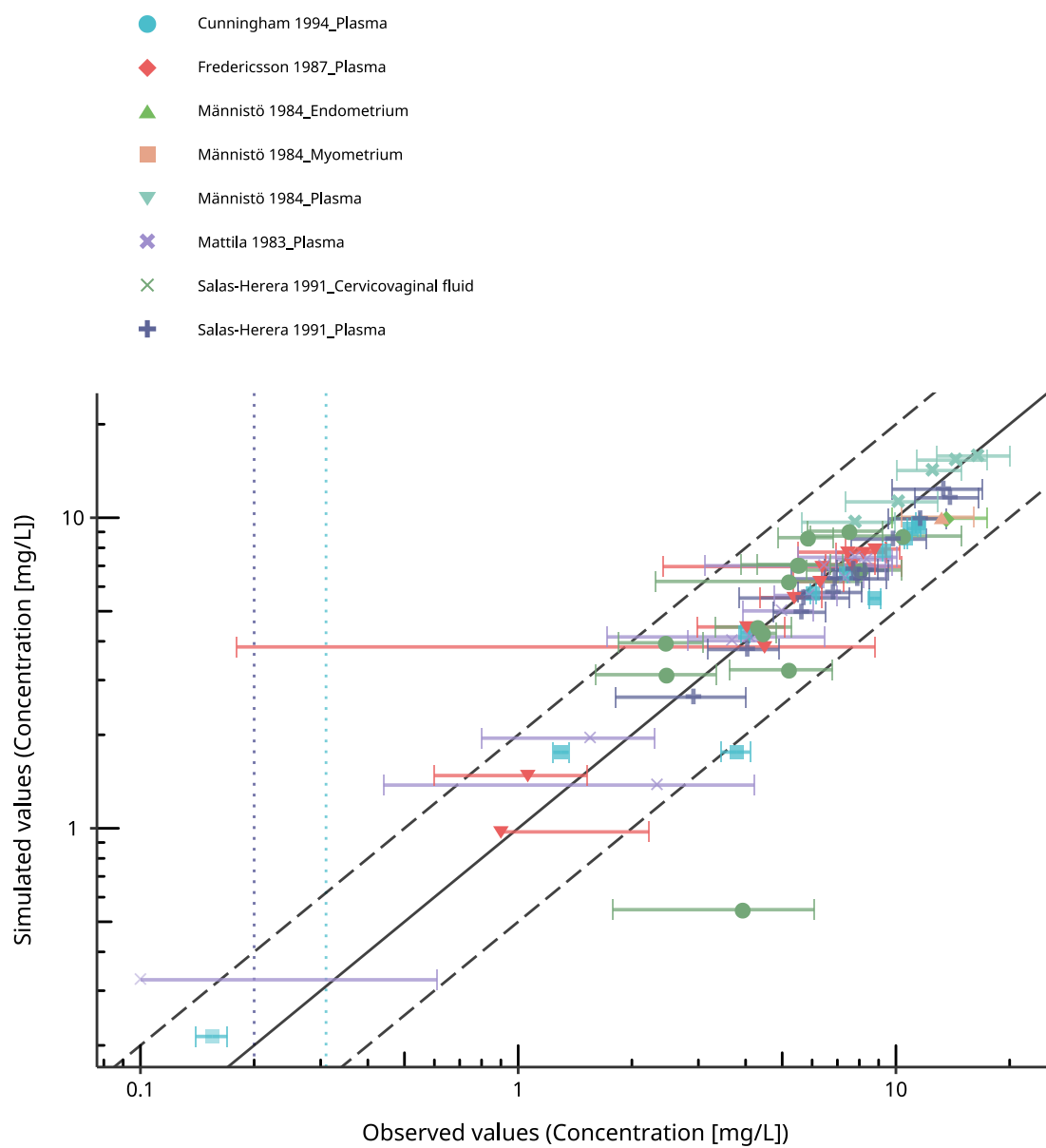


Figure 8: Predicted versus observed plot for intravenous administration of metronidazole.

Metronidazole vaginal administration

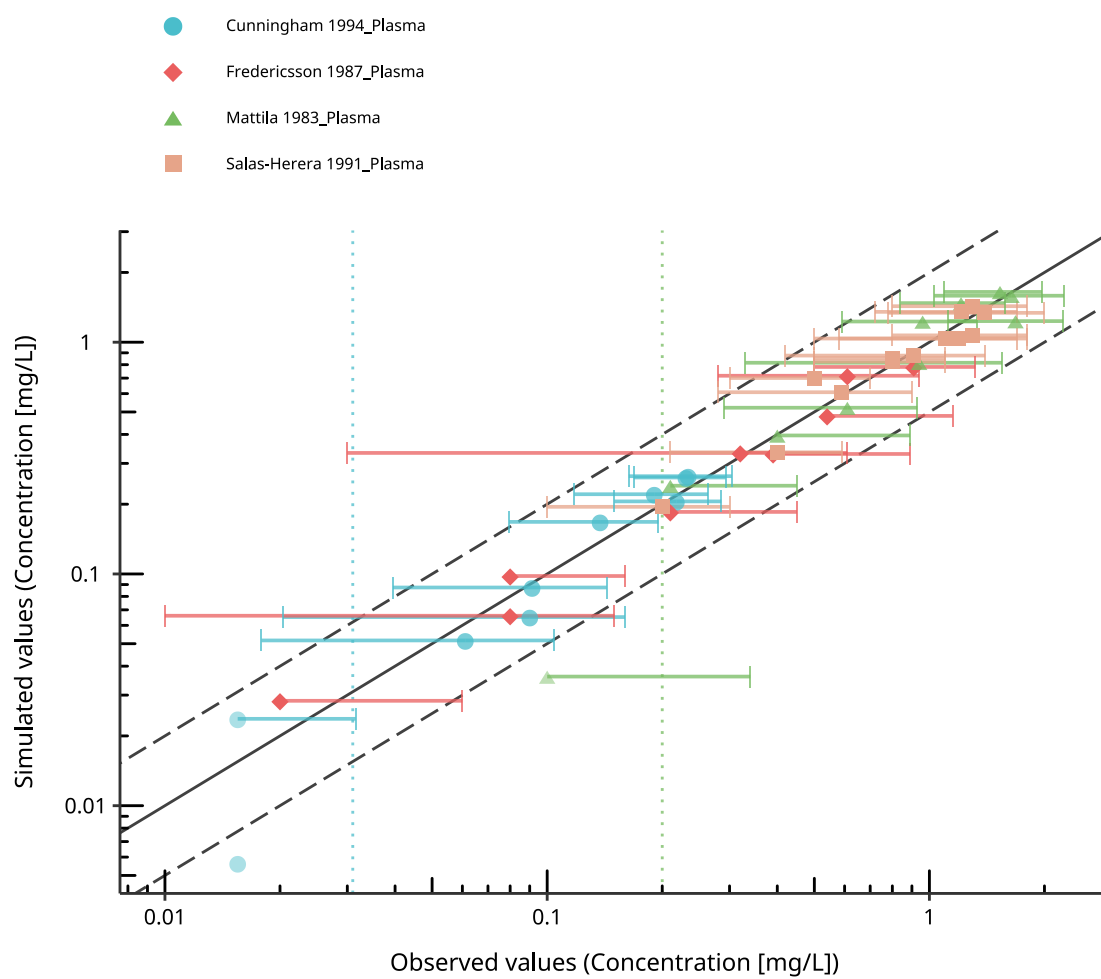


Figure 9: Predicted versus observed plot for vaginal administration of metronidazole.

5 References

References

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